Normal Calculations with a Z-Table

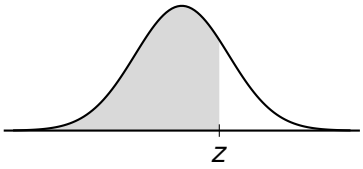
Name _____

The empirical rule only works for whole SDs (1, 2, or 3). A **z-table** works for any z-score.

The **z-table** gives the area to the **left** of z. That is the proportion of the data **below** z.

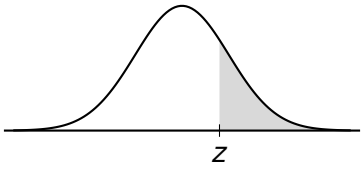
Reading the table: Find $z = 1.33$. Go down to the row **1.3**. Go across to the column **.03**. The area is **.9082**.

Below z



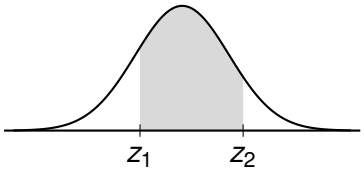
use the table

Above z



$1 -$ table

Between

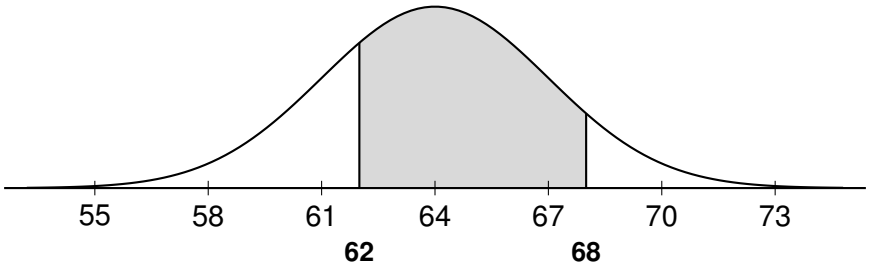


table(z_2) $-$ table(z_1)

Steps

1. Draw the curve. Shade the part the question asks about.
2. Find each z-score: $z = \dfrac{\text{value} - \text{mean}}{\text{SD}}$. Round to 2 decimal places.
3. Look up each z-score in the table.
4. Below: use the table. Above: 1$-$ table. Between: subtract.

Example: Heights of adult women: mean 64 inches, SD 3 inches.



Height of adult women (inches)

Shorter than 68 inches	$z = \frac{68-64}{3} = 1.33$ table: .9082 **90.82%**
Taller than 68 inches	$1 - .9082 = .0918$ **9.18%**
From 62 to 68 inches	$z = \frac{62-64}{3} = -0.67$ table: .2514 $.9082 - .2514 = .6568$ **65.68%**

Use your z-table sheet for the problems.

Work with a partner. Compare answers when you finish each problem.

1. Use the z-table. Find the area **below** each z-score.

   (a) $z = 1.25$

   (b) $z = -0.84$

   (c) $z = 2.00$

   (d) $z = -1.96$

2. Find the area **above** each z-score.

   (a) $z = 0.50$

   (b) $z = -1.20$

3. Find the area **between** the two z-scores.

   (a) $z = -1$ and $z = 1$

   (b) $z = 0.5$ and $z = 1.5$

4. The empirical rule says about 68% of the data is within 1 SD of the mean. Your answer to 3(a) is .6826. Do these agree? Explain.

5. Commute times to a school have a normal distribution with a mean of 25 minutes and an SD of 5 minutes. What percent of students have a commute:

   (a) less than 30 minutes?

   (b) more than 33 minutes?

   (c) from 20 to 28 minutes?

6. Apples from an orchard have a mean weight of 150 grams and an SD of 10 grams. The weights have a normal distribution.

   (a) The store sells apples over 165 grams as “extra large.” What percent of apples are extra large?

   (b) Apples under 138 grams are used for juice. What percent go to juice?